

AMAN RAY

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🌐 LinkedIn — 🐙 GitHub — 🤖 Hugging Face — 🏆 Kaggle

SUMMARY

M.Sc. Data Science student at Chennai Mathematical Institute with a strong mathematical foundation and hands-on experience in Speech AI, Large Language Models, transformer-based architectures and machine learning. AI Intern at Coriolis Technologies, working on Hindi Text-to-Speech systems using PyTorch, Hugging Face, PEFT/LoRA and Weights & Biases. Experienced in designing end-to-end AI pipelines spanning dataset curation, model fine-tuning, evaluation, deployment and reliability-focused verification.

TECHNICAL SKILLS

- **Programming:** Python, SQL, C
- **Deep Learning:** PyTorch, Hugging Face Transformers, PEFT, LoRA, CUDA
- **Machine Learning:** Scikit-Learn, XGBoost, Decision Trees, RandomizedSearchCV, Model Evaluation
- **Speech AI:** Text-to-Speech, Whisper, TorchAudio, Librosa, WER/CER Evaluation, Speaker Adaptation
- **Tools:** Git, GitHub, GitLab, Docker, Streamlit, Weights & Biases

EXPERIENCE

AI Intern — Coriolis Technologies

May 2026 – Jul 2026

- Fine-tuned the transformer-based Orpheus Text-to-Speech model for high-quality Hindi speech synthesis using PyTorch, Hugging Face and PEFT/LoRA.
- Built automated evaluation workflows using Whisper ASR and WER/CER metrics to benchmark synthesized speech.
- Designed a Local Outlier Factor (LOF) based selective verification pipeline to improve inference reliability for deployment.
- Collaborated on experimentation, optimisation and deployment-oriented improvements for an Edtech App.

Placement Committee Member — Chennai Mathematical Institute

Jun 2026 – Present

- Coordinate recruiter outreach, placement activities and communication between students, faculty and industry partners.

PROJECTS

Fine-tuning Orpheus TTS for Hindi Speech Synthesis

- Built an end-to-end Hindi speech synthesis pipeline by fine-tuning the transformer-based Orpheus TTS model on 75+ hours of curated Hindi speech.
- Developed multilingual metadata generation, preprocessing and dataset curation pipelines for large-scale speech training.
- Designed an automated evaluation framework over 500+ synthesized samples using Whisper ASR.
- Achieved a 64.3% reduction in Mean WER and an 80.6% reduction in Mean CER while evaluating multiple speakers and inference stages.

Speech-Based AI Soft Skills Analyzer

Live Demo — GitHub

- Built an AI-powered communication assessment platform integrating Whisper, Librosa and Llama 3.1.
- Developed an interactive Streamlit application generating structured reports evaluating clarity, confidence, engagement and pacing.

Bank Direct Marketing Optimization

GitHub

- Built an end-to-end machine learning pipeline using Scikit-Learn Pipelines, ColumnTransformer, SMOTE and Decision Trees.
- Optimized model performance using RandomizedSearchCV to improve customer targeting.

EDUCATION

Chennai Mathematical Institute

2025 – Present

M.Sc. Data Science

CGPA: 8.75/10

Relevant Coursework: Machine Learning, Probability & Statistics, Distributed Computing, Algorithms, Data Mining, Linear Algebra.

Banaras Hindu University

2021 – 2024

B.Sc. (Hons.) Mathematics

CGPA: 7.73/10

Relevant Coursework: Real Analysis, Linear Algebra, Abstract Algebra, Numerical Analysis, Operations Research, Programming in C.

ACADEMIC RESEARCH

Technical Review: Normalized Cuts and Image Segmentation

- Conducted an in-depth study of the Shi & Malik IEEE TPAMI paper on spectral graph theory and graph partitioning.
- Authored a 15-page technical report covering graph Laplacians, generalized eigenvalue formulations and Normalized Cuts.
- Implemented the complete algorithm in Python and presented the work as graduate coursework.